Text Mining and Sentiment Analysis Applied to Youth Soccer Training: A Case Study on the ASO Cernusco 2014

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Abstract

This report presents the application of Natural Language Processing (NLP) and text mining techniques to analyze my personal training logs as the Head Coach of the ASO Cernusco youth soccer team (Under-12 category, born in 2014). Although coaching is a volunteer activity, my approach to training and team management is deeply systematic. By utilizing Transformer models (BERT) for sentiment analysis and text-feature extraction techniques, it was possible to quantify the emotional trend of the squad over time and monitor the evolution of the tactical programming during the crucial transition from 7-a-side to 11-a-side soccer. The results demonstrate how a rigorous, data-driven approach can provide objective support even in grassroots sports contexts.

1 Introduction

In modern youth soccer, data collection is often limited to physical metrics or match events. However, unstructured qualitative data, such as post-session reports, contain invaluable insights regarding the psychological, disciplinary, and technical-tactical state of the team.

This season marked a significant milestone both for myself and the squad. For the players (born in 2014), it represented the critical and challenging transition from 7-a-side to 11-a-side soccer. Managing this shift entails a heavy responsibility, as it requires players to develop a fundamentally new spatial awareness, understanding concepts such as width (*ampiezza*), depth (*profondità*), the offside rule (*fuorigioco*), and organized pressing. To facilitate this cognitive and tactical leap, I progressively introduced constraints during practice—drawing upon the pedagogical principles of the constraints-led approach [1]—such as touch limits, the third-man rule (*regola del terzo uomo*), and point-based rewards for effective ball circulation (*giro palla*).

Concurrently, this was my first experience as a Head Coach, having previously served as an Assistant Coach for the 2011 age group for five years. To manage the weight of this new responsibility, ensure methodological rigor, and track the effectiveness of my didactics, I decided to systematically document and review every training session. This study aims to structure this text database, applying advanced NLP techniques to extract latent patterns and evaluate the training periodization.

2 Theoretical Background and Methodology

The textual corpus consists of my personal coaching notes spanning exactly three months, from the beginning of September to the beginning of December 2025. The schedule comprised two training sessions per week, with each session lasting 1.5 hours.

2.1 Training Structure and Pedagogical Focus

The ASO Cernusco 2014 team competes within the CSI (*Centro Sportivo Italiano*) circuit, a prominent grassroots and amateur sports organization. Given the inclusive nature of this context, players often join with varying levels of foundational soccer skills. To bridge these gaps, my methodological approach heavily prioritizes individual technique and ball mastery as well as an accurate first touch. To ensure systematic development, every training session follows a rigorous five-step structure:

1. **Activation:** A warm-up drill integrating technical aspects and funny components.
2. **Analytical Phase:** Drills focus on the repetition of fundamental technical gestures, paying attention to technical execution and agility.
3. **Situational Phase:** Application of skills under active, match-like pressure, with some element of individual tactic.
4. **Themed Games / SSG:** Small-Sided Games designed to trigger specific tactical behaviors.
5. **Final Match:** Unrestricted free play to consolidate learning in a competitive scenario.

The analytical pipeline to process the logs of these sessions was structured into three main phases: Text Preprocessing, Sentiment Analysis via Transformers, and Information Extraction.

2.2 Text Preprocessing and Tokenization

Raw textual data is inherently noisy. The initial phase involved standardizing the text to ensure computational efficiency. The text was lowercased, and punctuation was

removed. Subsequently, a custom stopword removal process was applied. Standard Italian stopwords were integrated with a domain-specific dictionary containing informal abbreviations used during note-taking (e.g., *vs, x, po*). This ensured that the subsequent frequency analysis focused solely on semantically rich terms.

2.3 Sentiment Analysis via BERT

Traditional lexicon-based sentiment analysis often fails to capture the nuances of natural language, as it evaluates words in isolation. To overcome this, I employed a Transformer-based architecture, specifically the pre-trained `neuraly/bert-base-italian-cased-sentiment` model.

BERT captures bidirectional context via a self-attention mechanism, computing scaled dot-products across Query (Q), Key (K), and Value (V) matrices (where d_k is the key dimension):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

To predict the sentiment, the model evaluates the final hidden state of the special `[CLS]` (classification) token, which aggregates the entire sequence's meaning. The final *Sentiment Score* ($S \in [-1, 1]$) is derived as follows:

$$S = \begin{cases} +P(\text{Positive}) & \text{if label is Positive} \\ -P(\text{Negative}) & \text{if label is Negative} \end{cases} \tag{2}$$

2.4 Information Extraction via Regex

To monitor the tactical progression tailored for the 11-a-side transition, a custom ontology of soccer concepts was defined. Since a single concept can be expressed through various synonyms (e.g., *3 uomo* or *terzo uomo*), Regular Expressions (Regex) were utilized for advanced pattern matching, generating accurate frequencies per training cycle.

3 Results

3.1 Longitudinal Sentiment Analysis

As shown in Figure 1, the model plotted the *Sentiment Score* chronologically across the 3-month period. This timeline captures the peaks of high engagement and the valleys associated with drops in attention or disciplinary interventions. This longitudinal view is crucial to understand how 11-year-olds respond to the increased intensity of the sessions.

Specifically, the plotted data reveals a highly responsive, cyclical behavioral pattern. Sharp declines in the *Sentiment Score* are consistently linked to specific disciplinary anomalies (e.g., lack of focus, locker-room distractions, or interpersonal conflicts among players). Notably, these "valleys" are almost universally followed by an immediate, sharp positive rebound in the subsequent training session. This quantifiable rebound provides empirical validation for the effectiveness of the coach's firm disciplinary interventions which successfully "reset" the group's cognitive focus, instantly restoring high training intensity and structural order.

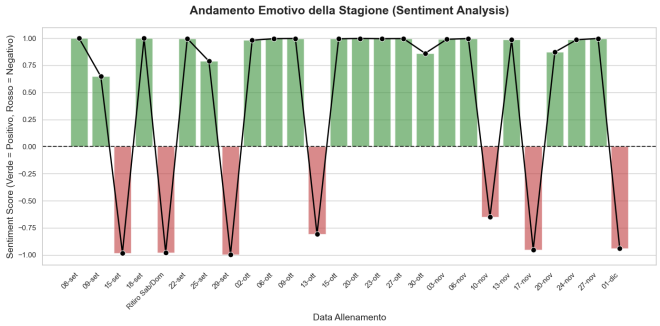

Figure 1: *Trend of the Sentiment Score over the sessions.*

3.2 Lexical Frequencies (Wordcloud)

To understand my primary instructional focus, a visual representation of lexical frequencies was generated (Figure 2). The dominant terms reflect a strong emphasis on "intensità" (intensity), cognitive development ("testa"/focus), and preferred practical drills to teach spacing, such as the "partitella" and the "torello" (rondo).

Furthermore, the high recurrence of evaluative and action-oriented words like "migliorare" (to improve) and "bene" (well/good) highlights a continuous loop of assessment and positive reinforcement, which is vital in a grassroots environment. The prominence of foundational technical concepts—such as "passaggi" (passing), "stop" (first touch), and "1v1" situations—directly mirrors the pedagogical commitment to bridging individual skill gaps during the analytical phases of the training sessions.

Figure 2: *Wordcloud of the coach's post-training notes. Font size is proportional to the term's frequency.*

3.3 Tactical and Educational Evolution

The Regex-based ontology tracked the presence of key tactical concepts within the reports, mapped across the five microcycles.

The heatmap (Figure 3) highlights the clear educational progression necessary for the 11-a-side transition.

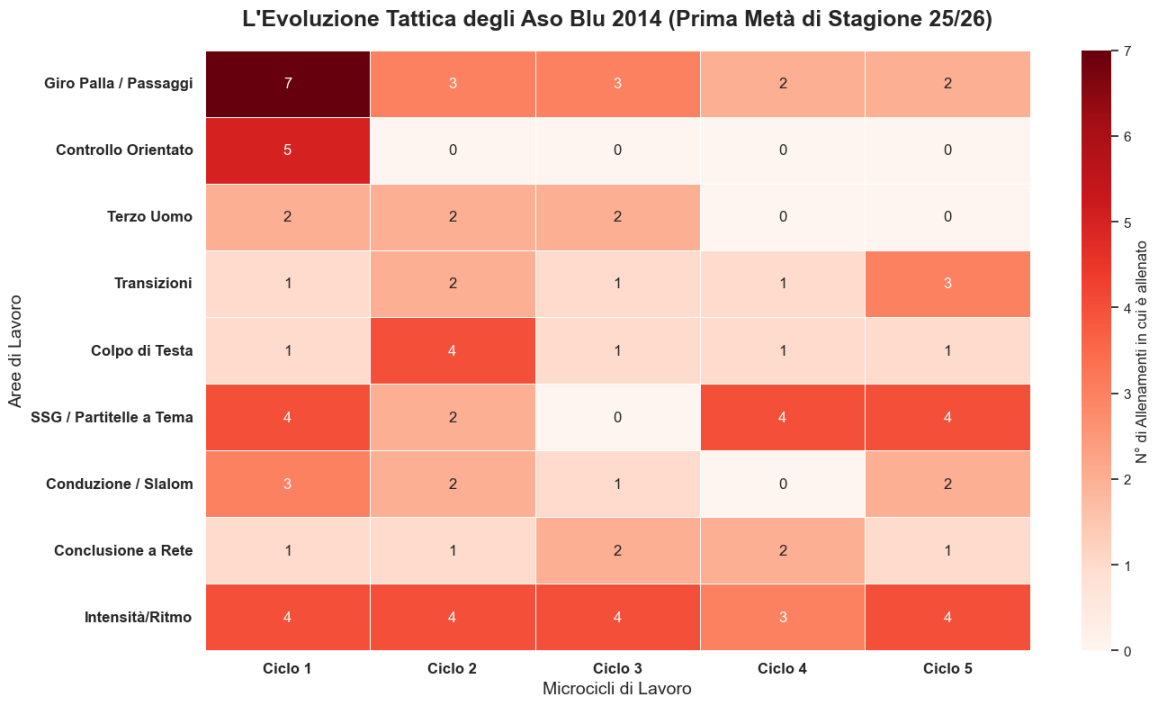

Figure 3: *Tactical evolution heatmap during the training microcycles.*

Cycle 1 was characterized by strong technical consolidation to handle larger spaces. The focus then shifted to specific technical gestures (*Colpo di Testa* in Cycle 2) and situations of systemic complexity, emphasizing the use of Small-Sided Games (SSG) and the training of transitions.

4 Conclusion

Applying advanced NLP techniques to a personal training diary proved to be a highly effective practice for a first-year Head Coach. The integration of Transformer models like BERT [2] allowed for a nuanced understanding of the squad's psychological dynamics during a delicate developmental phase. The longitudinal sentiment analysis successfully demonstrated how 11-year-olds respond to the cognitive load of a full-sized pitch and validated the efficacy of structural disciplinary interventions.

Concurrently, regex-based text mining provided objective validation of the tactical progression towards the 11-a-side format. The data confirmed the adherence to the initial pedagogical goals: prioritizing individual ball mastery before shifting to complex spatial paradigms trained through Small-Sided Games, a methodology widely supported in modern sports science for enhancing youth decision-making [3].

Ultimately, this case study demonstrates that advanced analytics are not exclusively reserved for professional, high-budget environments. By leveraging open-source algorithms and systematic documentation, it is possible to extract actionable insights even in a grassroots, volunteer-based context like the CSI circuit. This data-driven approach provides a solid, objective foundation for programming the second half of the season, effectively bridging the gap between grassroots coaching and academic sports analytics.

References

[1] I. Renshaw, K. Davids, D. Newcombe, and W. Roberts. *The Constraints-Led Approach: Principles for Sports Coaching and Practice Design.* Routledge (2019).

[2] J. Devlin, M. Chang, K. Lee, and K. Toutanova. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.* (2018).

[3] A. Hill-Haas, B. Dawson, F. Impellizzeri, and A. Coutts. *Physiology of small-sided games training in football: a systematic review.* Sports Medicine, 41(3), 199-220 (2011).